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Linear System Theory — Final Exam Cheatsheet

Lectures 2–23 | EE 545

PART 1: VECTOR FUNDAMENTALS

Combinations

Type	Formula	Constraint
Linear	$\sum \alpha_i x_i$	None
Affine	$\sum \alpha_i x_i$	$\sum \alpha_i = 1$
Convex	$\sum \alpha_i x_i$	$\sum \alpha_i = 1, \alpha_i \geq 0$

- **Convex set:** for any $x, y \in S$ and $t \in [0, 1]$, $tx + (1 - t)y \in S$.
- **Convex hull:** all convex combinations of points in a set.
- **Affine hull:** all affine combinations. Two points $\rightarrow$ line through them.

Hyperplanes and Half-Spaces

- **Hyperplane:** $\{x : a^T x = b\}$. In 2D: a line. Normal vector a .
 - **Half-space:** $\{x : a^T x \geq b\}$ or $\{x : a^T x \leq b\}$.
 - A neuron is a half-space classifier.
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PART 2: NORMS AND INNER PRODUCTS

Norm Axioms

1. $\|x\| \geq 0$; $\|x\| = 0 \iff x = 0$
2. $\|\alpha x\| = |\alpha| \|x\|$
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)

Vector p -Norms (on $\mathbb{C}^n$)

$$\|x\|_p = \left(\sum_k |x_k|^p \right)^{1/p}, \quad \|x\|_\infty = \max_k |x_k|$$

- ℓ_1 : taxicab/Manhattan; promotes sparsity.
- ℓ_2 : Euclidean; $\|x\|_2 = \sqrt{x^* x}$.
- ℓ_∞ : Chebyshev; worst-case error.

Inner Products

Real: $\langle x, y \rangle = x^T y$. **Complex:** $\langle x, y \rangle = y^* x$ (linear in first, conjugate-linear in second).

Properties: conjugate symmetry $\langle x, y \rangle = \overline{\langle y, x \rangle}$; positive definite $\langle x, x \rangle > 0$ for $x \neq 0$.

Cauchy-Schwarz: $|\langle x, y \rangle| \leq \|x\| \|y\|$

Angle: $\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|}$. **Orthogonal:** $x \perp y \iff \langle x, y \rangle = 0$.

Weighted 2-Norm

$\|x\|_W = \sqrt{x^* W x}$ for PD W . Eigenvectors of W define weighted directions; eigenvalues are the weights.

PART 3: VECTOR SPACES AND SUBSPACES

Definitions

- **Vector space:** set with scalar field, vector addition, scalar multiplication satisfying axioms.
- **Subspace:** subset closed under addition and scalar multiplication (contains zero).
- **Span:** all linear combinations of a set.
- **Linear independence:** $\sum \alpha_i x_i = 0 \implies \alpha_i = 0$.
- **Basis:** linearly independent spanning set.
- **Dimension:** number of vectors in any basis.

Four Fundamental Subspaces of A ($m \times n$, rank r)

Subspace	Definition	Ambient Space	Dimension
Column space $\mathcal{R}(A)$	$\{Ax : x \in \mathbb{C}^n\} = \text{span of columns}$	$\mathbb{C}^m$	r
Row space $\mathcal{R}(A^*)$	span of rows = $\mathcal{R}(A^T)$	$\mathbb{C}^n$	r
Null space $\mathcal{N}(A)$	$\{x : Ax = 0\}$	$\mathbb{C}^n$	$n - r$
Left null space $\mathcal{N}(A^*)$	$\{z : A^* z = 0\}$	$\mathbb{C}^m$	$m - r$

Orthogonality: $\mathcal{N}(A) \perp \mathcal{R}(A^*)$ and $\mathcal{N}(A^*) \perp \mathcal{R}(A)$.

Direct sums: $\mathbb{C}^n = \mathcal{R}(A^*) \oplus \mathcal{N}(A)$; $\mathbb{C}^m = \mathcal{R}(A) \oplus \mathcal{N}(A^*)$.

PART 4: $Ax = b$ — EXISTENCE AND UNIQUENESS

Condition	Requirement	Shape Needed
Solution exists	$b \in \mathcal{R}(A)$	—
Exists for ALL b	$\mathcal{R}(A) = \mathbb{C}^m$	Square or fat ($n \geq m$)
Unique (when exists)	$\mathcal{N}(A) = \{0\}$	Square or tall ($m \geq n$)
Both for all b	Full rank, square	Square only

- **All solutions:** $\hat{x} + \mathcal{N}(A)$ (one particular solution + null space).
- **Right inverse:** $AD = I_m$ exists iff $\mathcal{R}(A) = \mathbb{C}^m$ (full row rank).
- **Left inverse:** $CA = I_n$ exists iff $\mathcal{N}(A) = \{0\}$ (full column rank).

Rank: $\text{rank}(A) = \dim \mathcal{R}(A) = \dim \mathcal{N}(A^*)$. Row rank = column rank.

PART 5: MATRIX FACTORIZATIONS

Factorization	Form	Conditions	Purpose
PLU	$P^T AP = LU$	Any square	Gaussian elimination
Eigenvalue	$A = T\Lambda T^{-1}$	Diagonalizable	System analysis
Schur	$A = UTU^*$	Any square	Triangularization
QR	$A = QR$	Any matrix	Least squares
Cholesky	$A = LL^*$	$A \succ 0$	PD system solving
SVD	$A = U\Sigma V^*$	Any matrix	All-purpose analysis

Eigenvalues and Diagonalization

- **Eigenvalue equation:** $Ax = \lambda x$, $x \neq 0$.
 - **Characteristic polynomial:** $p(\lambda) = \det(\lambda I - A)$. Roots = eigenvalues.
 - **Eigenspace:** $\mathcal{N}(A - \lambda I)$.
 - **Diagonalizable:** $A = T\Lambda T^{-1}$ iff A has n linearly independent eigenvectors.
 - **Similarity:** $T^{-1}AT$ has the same eigenvalues as A .
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PART 6: ORTHOGONAL PROJECTION

Orthogonal Projection of x onto Subspace V

- $p \in V$ and $x - p \perp V$.
- Minimizes $\|x - v\|$ over all $v \in V$.

Projection Matrix

If Q has orthonormal columns spanning V :

$$P = QQ^*, \quad P^2 = P, \quad P^* = P.$$

If columns are not orthonormal: $P = A(A^*A)^{-1}A^*$.

PART 7: SPECIAL MATRIX CLASSES

Hierarchy

All matrices

- └ Diagonalizable ($T^{-1}AT = \Lambda$)
 - └ Normal ($A^*A = AA^*$) = Unitarily diagonalizable ($A = U\Lambda U^*$)
 - └ Hermitian ($A^* = A$): eigenvalues real
 - └ Positive definite/semidefinite/indefinite (by sign of eigenvalues)
 - └ Unitary ($U^*U = I$): eigenvalues on unit circle
 - └ Skew-Hermitian ($A^* = -A$): eigenvalues on imaginary axis

Key Properties

Matrix	Condition	Eigenvalues	x^*Ax
Hermitian	$A^* = A$	Real	Real
Unitary	$U^*U = I$	$ \lambda = 1$	Preserves norm
Normal	$A^*A = AA^*$	Any	—
PD Hermitian	All $\lambda > 0$	Positive	> 0 for $x \neq 0$
PSD Hermitian	All $\lambda \geq 0$	Nonneg	≥ 0

Unitary Matrices

- $U^*U = UU^* = I$; $U^{-1} = U^*$.
- Columns and rows form orthonormal bases.
- $\|Ux\|_2 = \|x\|_2$; $\langle Ux, Uy \rangle = \langle x, y \rangle$.
- Eigenvalues on unit circle: $|\lambda| = 1$ (proof: $\|Ux\| = \|\lambda x\| = |\lambda|\|x\|$).
- Eigenspaces for distinct eigenvalues are orthogonal.

Hermitian Matrices

- $A^* = A$; real version: symmetric.
- $x^*Ax \in \mathbb{R}$ for all x : proof $(x^*Ax)^* = x^*A^*x = x^*Ax$.
- Eigenvalues real: from $x^*Ax = \lambda x^*x$ with $x^*x > 0$.
- Eigenspaces for distinct eigenvalues orthogonal.

Normal Matrices

- $A^*A = AA^*$ iff unitarily diagonalizable iff orthonormal eigenbasis exists.
- Trace identity: $\|A\|_F^2 = \sum_i |\lambda_i|^2$ (for normal matrices).

Schur Factorization

Every square complex A has $A = UTU^*$ where U unitary and T upper triangular.

Proof idea: Choose eigenvector, extend to ONB, get block triangular, induct.

PART 8: POSITIVE DEFINITE MATRICES

Equivalences (for Hermitian A)

$$A \succ 0 \iff \text{all eigenvalues} > 0 \iff x^*Ax > 0 \ \forall x \neq 0 \iff A = SS^* \text{ for invertible } S.$$

Sign Classes

Class	Eigenvalues	Quadratic Form
PD	All > 0	$x^*Ax > 0$
PSD	All ≥ 0	$x^*Ax \geq 0$
ND	All < 0	$x^*Ax < 0$
NSD	All ≤ 0	$x^*Ax \leq 0$
Indefinite	Mixed	Both signs possible

Inertia and Sylvester's Law

- **Inertia:** $(n_+, n_-, n_0) = (\# \text{ positive}, \# \text{ negative}, \# \text{ zero eigenvalues})$.
- **Star-congruence:** $A = SBS^*$ for invertible S preserves inertia.
- **Sylvester's law:** $A \stackrel{*}{\sim} B \iff \text{inertia}(A) = \text{inertia}(B)$.
- **Corollary:** $A \succ 0 \iff A \stackrel{*}{\sim} I \iff A = SS^*$.

Diagonal Dominance (Sufficient, NOT Necessary)

$a_{ii} > \sum_{j \neq i} |a_{ij}|$ for all i implies $A \succ 0$.

Cholesky Factorization

$A \succ 0 \implies \exists$ unique lower triangular L with positive diagonal: $A = LL^*$.

Proof sketch: block-partition, Schur complement is PD by star-congruence, induct.

Solving $Px = y$ (PD P): Factor $P = LL^*$, then solve $Lz = y$, then $L^*x = z$.

Gram Matrices

For any Z : Z^*Z is always PSD. If Z has full column rank, $Z^*Z \succ 0$.

PART 9: LEAST SQUARES

Problem: A is tall full-rank ($m > n$, rank n). No exact solution for general $b \in \mathbb{R}^m$.

$$\min_x \|Ax - b\|_2^2$$

Normal equations: $A^*A\hat{x} = A^*b$. Since $A^*A \succ 0$:

$$\hat{x} = (A^*A)^{-1}A^*b.$$

Pseudo-inverse: $A^\dagger = (A^*A)^{-1}A^*$ (left inverse of A).

Orthogonality condition: $b - A\hat{x} \perp \mathcal{R}(A)$.

Numerically: Use QR or SVD. Avoid forming A^*A (squares condition number).

PART 10: QR FACTORIZATION

$A = QR$ where Q has orthonormal columns and R is upper triangular.

- **Gram-Schmidt:** Orthonormalize columns with causal constraint $\rightarrow$ upper triangular R .
- **Modified Gram-Schmidt:** Numerically superior; subtract q_k component from all remaining vectors as soon as q_k is found.
- **Full QR:** Q is square unitary; R is rectangular.
- **Reduced QR:** Q has same number of columns as A ; R is square.

Causal constraint: $\text{span}\{q_1, \dots, q_k\} = \text{span}\{a_1, \dots, a_k\}$.

PART 11: BLOCK MATRIX TECHNIQUES

Schur Complement

For $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ with A_{11} invertible:

$$S_{11} = A_{22} - A_{21}A_{11}^{-1}A_{12} \quad (\text{Schur complement of } A_{11}).$$

$$\det(A) = \det(A_{11}) \det(S_{11}).$$

PD criterion: $A \succ 0 \iff A_{11} \succ 0$ and $S_{11} \succ 0$.

Inertia additivity (Hermitian A): $\text{inertia}(A) = \text{inertia}(A_{11}) + \text{inertia}(S_{11})$.

Matrix Inversion Lemma (Woodbury Identity)

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}.$$

Application: rank-one updates of inverse in RLS algorithm.

PART 12: MATRIX NORMS

Frobenius Norm

$$\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2} = \sqrt{\text{tr}(A^*A)} = \sqrt{\sum_k \sigma_k^2}.$$

Induced Norms

$$\|A\|_{q,p} = \sup_{\|x\|_p=1} \|Ax\|_q.$$

Norm	Formula	Computation
$\ A\ _{1,1}$	ℓ_1 in, ℓ_1 out	Max column ℓ_1 -norm

Norm	Formula	Computation
$\ A\ _{\infty, \infty}$	ℓ_{∞} in, ℓ_{∞} out	Max row ℓ_1 -norm
$\ A\ _{2,2} = \ A\ _2 = \sigma_1$	ℓ_2 in, ℓ_2 out	Largest singular value

Geometric view: Unit ℓ_2 sphere $\rightarrow$ ellipsoid; longest semi-axis = σ_1 = induced 2-norm.

Schatten p -Norms

$$\|A\|_{S_p} = \|\sigma\|_p = \left(\sum_k \sigma_k^p \right)^{1/p}.$$

- $p = 1$: **nuclear norm** $\|A\|_* = \sum_k \sigma_k$. Convex relaxation of rank. - $p = 2$: Frobenius norm. - $p = \infty$: induced 2-2 norm = σ_1 .

PART 13: SINGULAR VALUE DECOMPOSITION

Definition

For any $m \times n$ matrix A of rank r :

$$A = U\Sigma V^*, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0.$$

- U : $m \times m$ unitary (left singular vectors as columns).
- V : $n \times n$ unitary (right singular vectors as columns).
- Σ : $m \times n$ real nonneg diagonal.

Outer product form: $A = \sum_{k=1}^r \sigma_k u_k v_k^*$.

SVD Reveals Everything

Property	From SVD
Rank	# nonzero σ_k
Column space	First r cols of U
Left null space	Last $m - r$ cols of U
Row space	First r cols of V
Null space	Last $n - r$ cols of V
Induced 2-norm	σ_1
Frobenius norm	$\sqrt{\sum \sigma_k^2}$

Connection to Eigenvalues

- Singular values of $A = \sqrt{\text{eigenvalues of } A^*A}$.
- Right singular vectors = eigenvectors of A^*A .
- Left singular vectors = eigenvectors of AA^* .

Best Rank- p Approximation (Eckart-Young)

$$A_p = \sum_{k=1}^p \sigma_k u_k v_k^*, \quad \|A - A_p\|_F = \sqrt{\sum_{k=p+1}^r \sigma_k^2}.$$

A_p is the closest rank- p matrix to A in both Frobenius and induced 2-norm.

PART 14: ABSTRACT INNER PRODUCT SPACES

Axioms for Inner Product $\langle \cdot, \cdot \rangle$

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (conjugate symmetry)
2. Linear in first argument
3. $\langle x, x \rangle \geq 0$; $= 0$ iff $x = 0$

Induced norm: $\|x\| = \sqrt{\langle x, x \rangle}$. **Hilbert space** = complete inner product space.

Examples

- $\mathbb{C}^n$: $\langle x, y \rangle = y^* x$.
 - $L^2([a, b])$: $\langle f, g \rangle = \int_a^b f(t) \overline{g(t)} dt$.
 - Matrices: $\langle A, B \rangle = \text{tr}(A^* B) \rightarrow$ Frobenius norm.
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KEY PROOF IDEAS (EXAM-READY)

1. **Ax=b solvable iff $b \in \mathcal{R}(A)$** : expand Ax as column combination.
 2. **All solutions** = $\hat{x} + \mathcal{N}(A)$: subtract two solutions $\rightarrow$ difference in null space.
 3. **$\mathcal{N}(A) \perp \mathcal{R}(A^*)$** : $Ax = 0$ means inner product with every row is 0.
 4. **Row rank = column rank**: map row-basis through A , show independence; reverse.
 5. **Basis change** $\rightarrow T^{-1}AT$: convert coordinates, apply, convert back.
 6. **$U^*U = I$ preserves norm**: $\|Ux\|^2 = x^* U^* U x = x^* x$.
 7. **Unitary eigenvalues on unit circle**: norm preservation $\rightarrow |\lambda| = 1$.
 8. **Hermitian eigenvalues real**: $x^* A x = \lambda x^* x$, left side real by $A = A^*$.
 9. **Hermitian eigenspaces orthogonal**: compare $\lambda y^* x$ and $\mu y^* x \rightarrow 0$.
 10. **Normal = unitarily diagonalizable**: Schur $\rightarrow T$ normal $\rightarrow T$ diagonal.
 11. **PD criterion via quadratic form**: $A = U\Lambda U^*$, write $x^* A x = \sum \lambda_i |z_i|^2$.
 12. **$Z^* Z \succeq 0$** : $x^* (Z^* Z) x = \|Zx\|^2 \geq 0$.
 13. **Cholesky Schur complement PD**: by star-congruence preserving inertia.
 14. **Projection $P = QQ^*$** : $Q^* Q = I$; $Px \in V$; $x - Px \perp V$.
 15. **Eckart-Young**: truncate SVD to first p terms = best rank- p approximation.
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COMMON MISTAKES TO AVOID

- $A > 0$ for matrices means PD (not elementwise positive).
- Orthogonal $\neq$ orthonormal: orthonormal also requires unit norms.
- PSD $\neq$ PD: zero eigenvalues allowed in PSD, not in PD.

- Diagonal dominance is **sufficient** for PD, not necessary.
- Similarity preserves eigenvalues; star-congruence preserves **inertia** (not eigenvalues).
- SVD uses **two** unitary matrices (not one like eigendecomposition).
- Frobenius norm $\neq$ induced 2-norm: $\|A\|_F = \sqrt{\sum \sigma_k^2}$ vs $\|A\|_2 = \sigma_1$.
- Nuclear norm promotes low rank; it does NOT equal rank.
- Gram-Schmidt with linear dependence: the dependent vector produces $\tilde{a}_k = 0$; rank $<$ n.
- Left null space $\neq$ null space: $\mathcal{N}(A^*)$ is in the **output** space; $\mathcal{N}(A)$ is in the **input** space.